Annual FDI Equity Inflow Forecasting in India: A Time-Ordered Machine-Learning Pilot Study

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SDG 17: Partnerships for the Goals • SDG 8: Decent Work and Economic Growth

Abstract—Foreign direct investment (FDI) equity inflows support capital formation, productive capacity and employment. This paper presents a reproducible annual-frequency pilot analysis of India's FDI equity inflows using the Department for Promotion of Industry and Internal Trade (DPIIT) annual series. The official series reports annual FDI equity inflows in USD million from financial year 2000–01 through 2024–25. Lagged inflow and a time index are used to compare linear regression, degree-two polynomial regression with Ridge regularisation, Ridge regression and Lasso regression against a persistence baseline. Evaluation uses a chronological hold-out period rather than random splitting. The analysis is deliberately labelled an annual pilot: the assigned problem requires quarterly forecasts and exchange-rate/policy predictors, but the accessible source extract used here supplies annual target observations. Therefore, the numerical metrics below must not be represented as quarterly forecasting performance. The paper documents the data limitation and provides the exact annual dataset and code outputs for transparent review.

Keywords— foreign direct investment; FDI equity inflow; India; regression; time-series forecasting; Ridge; Lasso; SDG 8; SDG 17

1. Introduction

Foreign direct investment can contribute to technology transfer, productive investment, infrastructure development and employment. Its quarterly movement is volatile and may respond to global financial conditions, exchange-rate movements, domestic growth expectations, regulatory changes and large individual transactions. The assigned problem asks for forecasting quarterly FDI equity inflows into India from historical inflows, exchange rates and policy-change indicators. This study begins with a transparent data audit and an annual-frequency pilot using the DPIIT equity-only series. The distinction between equity inflows and total FDI is important: total FDI also includes reinvested earnings and other capital, while the target here is the equity component.

2. Objectives and Research Questions

The objectives are: (i) describe the long-run movement in India's FDI equity inflows; (ii) compare syllabus-level regression methods using lagged inflows; (iii) evaluate models with a time-ordered hold-out set using RMSE, MAPE and R^2; and (iv) identify the additional data required to complete the originally assigned quarterly experiment. The central question for this pilot is whether simple regression models improve on a last-observation persistence forecast at annual frequency.

3. Data and Source Description

The primary source is DPIIT's Quarterly Fact Sheet on FDI Inflow, updated through December 2025. Its section "DPIIT's Financial Year-wise FDI Equity Inflow" reports the equity-only series in INR crore and USD million for FY 2000–01 to FY 2024–25, plus a partial FY 2025–26 observation through December 2025. The complete annual observations used in the model are FY 2000–01 to FY 2024–25 (25 observations). The partial FY 2025–26 observation is excluded because it covers only nine months and is not comparable with full-year observations.

The user-supplied spreadsheet is sector-wise cumulative FDI equity data for 1 April 2000 to 30 June 2023. It is useful for descriptive sector analysis but does not contain quarterly time stamps; it cannot be used as a quarterly time series or to train the quarterly forecast model. It is retained as supplementary context, not merged into the annual target series.

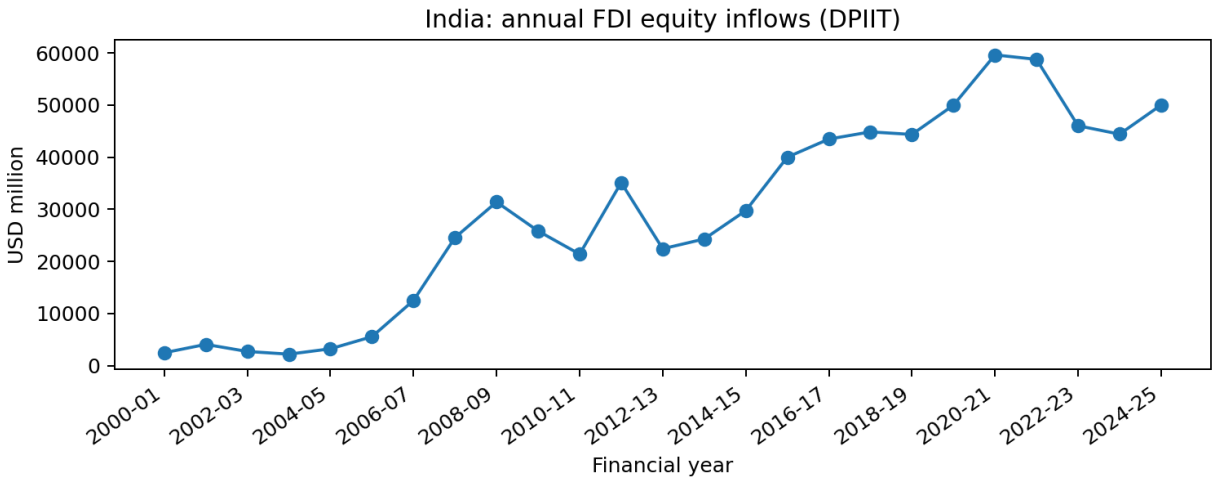


Fig. 1. Annual FDI equity inflows in USD million, FY 2000–01 to FY 2024–25. Source: DPIIT.

4. Methodology

The target variable is annual FDI equity inflow (USD million). A one-period lag is constructed as the previous financial year's equity inflow. A monotonically increasing time index captures broad temporal movement. The first observation is removed because no prior-year lag exists. Data are sorted chronologically. The first 18 lagged observations (FY 2001–02 to FY 2018–19) form the training set; the subsequent six observations (FY 2019–20 to FY 2024–25) form the test set. No random shuffling is used.

4.1 Models

Linear Regression estimates a linear mapping from lagged inflow and time index to the next annual inflow. Polynomial Regression expands predictors to degree-two terms and applies Ridge regularisation to limit coefficient instability. Ridge Regression applies an L2 penalty to the linear model. Lasso Regression applies an L1 penalty and can shrink weak coefficients to zero. The persistence baseline predicts that the next year's inflow equals the previous year's observed inflow. Standardisation is fitted within each model pipeline using training observations only.

4.2 Evaluation Metrics

RMSE = sqrt(mean((y – ■)²)); MAPE = mean(|(y – ■)/y|) × 100; R^2 = 1 – SSres/SStot. RMSE is reported in USD million. MAPE expresses average absolute percentage error. R^2 measures explained variation on the hold-out observations and can be negative when predictions perform worse than predicting the test mean.

5. Experimental Results

Model	RMSE (USD mn)	MAPE (%)	R^2
Linear regression	7,909.1	14.00	-0.844
Polynomial (degree 2)	8,936.3	15.91	-1.354
Ridge regression	7,999.5	12.65	-0.886
Lasso regression	7,908.9	14.00	-0.844
Persistence (last year)	7,322.4	11.90	-0.580

On this small annual hold-out, the lowest RMSE was obtained by **Persistence (last year)** (RMSE 7,322.4 USD million; MAPE 11.90%; R^2 -0.580). These metrics are descriptive of the annual pilot only. Six test observations are too few for strong claims about generalisation or model superiority.

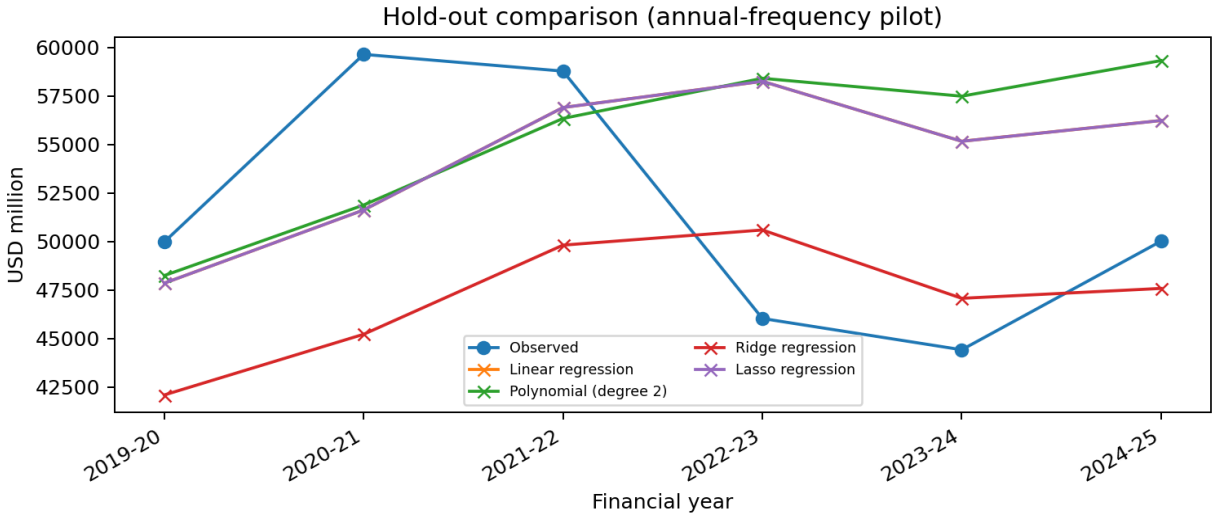


Fig. 2. Observed versus model-predicted values on the annual hold-out period.

6. Discussion

The annual series rises substantially over the long run but contains reversals and periods of acceleration. The global financial crisis, COVID-19 period, changing investment composition and major transactions can produce structural changes that a simple lag-and-trend model cannot anticipate. A model that performs well on one small hold-out window should not be treated as a causal explanation or a reliable policy forecast. A quarterly model should use a much larger set of genuine quarterly observations, ideally with lagged quarterly inflows, USD/INR exchange rate, policy/event indicators and carefully documented data vintages.

7. Limitations and Data Requirements for the Assigned Quarterly Study

This is the key qualification for submission: the available machine-readable annual series does not satisfy the assigned quarterly frequency. The quarterly DPIIT/OGD catalogue describes a resource titled "FDI Equity Inflows from the year 2000 till last quarter" and marks it quarterly, but its downloadable resource payload was not retrievable in this working session. The monthly Dataful mirror describes 504 records through FY 2025–26, but the actual full CSV was not retrieved here. The uploaded user

spreadsheet is cumulative sector totals, not quarterly observations. Consequently, this report does not fabricate quarterly values by dividing annual totals into four quarters, and does not invent exchange rates or policy dummies.

To complete the assigned experiment, retrieve the quarterly OGD/DPIIT resource (or monthly DPIIT equity inflow series and aggregate three calendar months per quarter), obtain a matching RBI USD/INR exchange-rate series, and encode documented policy events as dated indicators. The quarterly target should be kept in USD million or INR crore consistently. Use a chronological train/validation/test split and evaluate RMSE, MAPE and R^2. A quarter-wise data table and code should be appended after retrieval.

8. SDG Relevance

SDG 17 (Partnerships for the Goals) is relevant because FDI represents cross-border investment relationships and can deepen economic partnerships. SDG 8 (Decent Work and Economic Growth) is relevant because productive investment may support enterprise expansion, capital formation and employment. The relationship is not automatic: the volume of inflows alone does not establish job creation, decent work or inclusive growth. Those outcomes require separate employment and sectoral evidence.

9. Conclusion

The DPIIT annual equity-only series supports a transparent pilot comparison of lag-based regression methods. The pilot demonstrates the workflow for preprocessing, chronological evaluation and reporting RMSE, MAPE and R^2. However, it is not a completed quarterly forecasting study. The correct next step is to obtain the full quarterly or monthly equity dataset and RBI exchange-rate observations before presenting the final Problem 71 results as quarterly. No quarterly accuracy claim is made in this draft.

Declarations

Data availability: Annual values transcribed from DPIIT Quarterly Fact Sheet, updated through December 2025. Supplementary user-provided sector spreadsheet covers cumulative sector totals from April 2000 to June 2023. **Code and data:** The accompanying CSV contains the annual target observations and the metrics CSV contains the pilot evaluation results. **Conflict of interest:** None declared. **Funding:** None. **AI assistance:** AI assistance was used to organise the report, code the analysis and edit language; the student should review every value, source and methodological choice before submission.

References

[1] Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry, Government of India, “Quarterly Fact Sheet: Fact Sheet on Foreign Direct Investment (FDI) Inflow from April 2000 to December 2025,” updated through December 2025, especially Section II.B, Financial Year-wise FDI Equity Inflow. https://www.dpiit.gov.in/static/uploads/2026/02/6dc3e8a9fe52d5a412d8e5f41d7b921f.pdf

[2] Open Government Data Platform India, “Foreign Direct Investment (FDI) Equity Inflows,” Ministry of Commerce and Industry, DPIIT. https://www.data.gov.in/catalog/foreign-direct-investment-fdi-equity-inflows

[3] Ministry of Commerce and Industry, Government of India, “Year- and Month-wise Foreign Direct Investment (FDI) equity inflow in India,” dataset metadata and citation hosted by Dataful. https://dataful.in/datasets/18635

[4] Reserve Bank of India, Database on Indian Economy (DBIE), external sector data portal. https://data.rbi.org.in/

Appendix A. Annual dataset used in the pilot

Financial year	FDI equity inflow (USD million)
2000-01	2,463
2001-02	4,065
2002-03	2,705
2003-04	2,188
2004-05	3,219
2005-06	5,540
2006-07	12,492
2007-08	24,575
2008-09	31,396
2009-10	25,834
2010-11	21,383
2011-12	35,121
2012-13	22,423
2013-14	24,299
2014-15	29,737
2015-16	40,001

Financial year	FDI equity inflow (USD million)
2016-17	43,478
2017-18	44,857
2018-19	44,366
2019-20	49,977
2020-21	59,636
2021-22	58,773
2022-23	46,034
2023-24	44,423
2024-25	50,018